

Subgroup Performance

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can calculate subgroup results with sample sizes.

I can distinguish missing evidence from equal performance.

I can report differences without assigning causes from a table.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Overall metrics can conceal uneven behavior across meaningful conditions or user groups.



LEARNING TARGET Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the subgroup gap. What causal claim is not supported?

Builder name and experiment date:

Readiness for the connected explanation

What can a pooled error rate hide?

This week's end goal

By the end, I can explain subgroup performance using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → **Predict** → **Explain**

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use subgroup performance to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A single overall score can hide different experiences. Define relevant groups before evaluation and report counts as well as rates. This classroom exercise uses fictional sensor conditions, not personal identities. Real evaluations involving people require appropriate permissions, privacy and contextual expertise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines subgroup evaluation.

3. See how it works

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



IN

INPUT

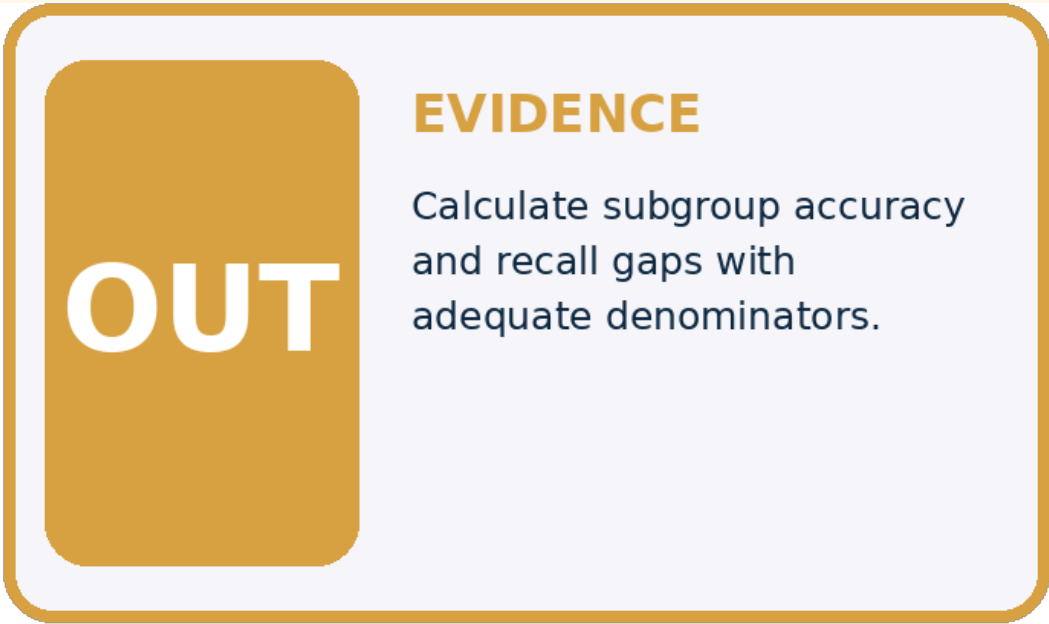
Calculate subgroup accuracy and recall gaps with adequate denominators.



ML

MODEL

Overall metrics can conceal uneven behavior across meaningful conditions or user groups.



OUT

EVIDENCE

Calculate subgroup accuracy and recall gaps with adequate denominators.

QUESTION Calculate subgroup accuracy and recall gaps with adequate denominators.

MODEL IDEA Overall metrics can conceal uneven behavior across meaningful conditions or user groups.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Daylight: 18/20 correct = 90%; evening: 10/20 = 50%. Accuracy gap = 90% – 50% = 40 percentage points.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

Sensor group A has 2 errors in 20 examples and group B has 3 in 5. Error rates are 10 and 60 percent; pooled error is 5/25=20 percent. The group difference is 50 percentage points. B has only five examples, so its rate is sensitive to each individual result.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Shade gains 5 new cases with no errors.

Prediction: Shade error becomes $3/10=30\%$; pooled error becomes $5/30\approx 16.7\%$.

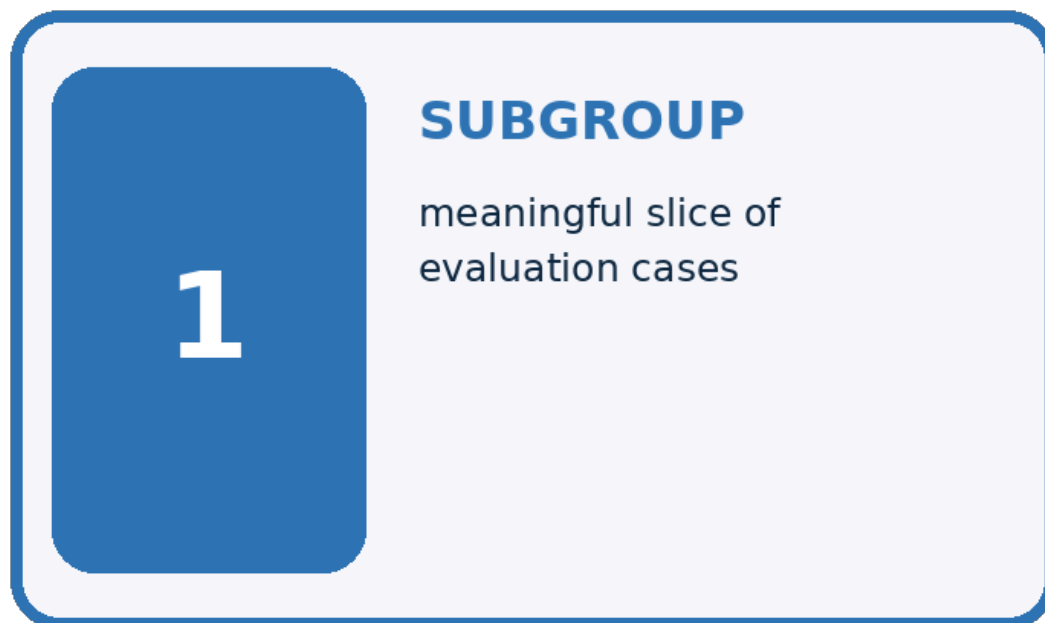
Reason: Extra evaluated cases change denominators; existing errors have not disappeared.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

PERFORMANCE GAP

difference between subgroup metrics

3

DENOMINATOR

number of cases behind a rate

4

INTERSECTION

cases belonging to multiple slices

Term · Precise meaning · My ML example

Term: subgroup

Precise meaning: meaningful slice of evaluation cases

My ML example: _____

Term: performance gap

Precise meaning: difference between subgroup metrics

My ML example: _____

Term: denominator

Precise meaning: number of cases behind a rate

My ML example: _____

Term: intersection

Precise meaning: cases belonging to multiple slices

My ML example: _____

MISCONCEPTION TO REPAIR A good overall result guarantees acceptable performance for every subgroup.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Subgroup

Meaning: A defined subset of the evaluated examples

Word: Disaggregation

Meaning: Reporting results separately for defined groups

Word: Coverage

Meaning: Which groups and conditions are represented

Word: Disparity

Meaning: An observed difference between group results

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 31 • WORKED EVIDENCE

- 1 Daylight: $18/20$ correct = 90%; evening: $10/20 = 50\%$.
- 2 Accuracy gap = $90\% - 50\% = 40$ percentage points.
- 3 Evening spotted recall is $4/10 = 40\%$, versus daylight $8/10 = 80\%$.
- 4 Inspect representation and image quality; report counts and uncertainty before repairing.

STEP 1 Daylight: $18/20$ correct = 90%; evening: $10/20 = 50\%$.

STEP 2 Accuracy gap = $90\% - 50\% = 40$ percentage points.

STEP 3 Evening spotted recall is $4/10 = 40\%$, versus daylight $8/10 = 80\%$.

STEP 4 Inspect representation and image quality; report counts and uncertainty before repairing.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Subgroup evaluation at a glance

Part • Evidence or action • Interpretation

Part: Group A

Evidence or action: 2 errors of 20

Interpretation: 10 percent

Part: Group B

Evidence or action: 3 errors of 5

Interpretation: 60 percent

Part: Pooled

Evidence or action: 5 errors of 25

Interpretation: 20 percent

Changing one B error to a correct result changes its rate from 60 to 40 percent. Small samples can be unstable without making the observed errors unimportant. A missing group has no measured rate, not zero error. Intersections of conditions may be even smaller.

How does correcting one B error affect its rate? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Subgroup Performance

Calculate subgroup and pooled error rates without letting the larger group hide a gap.

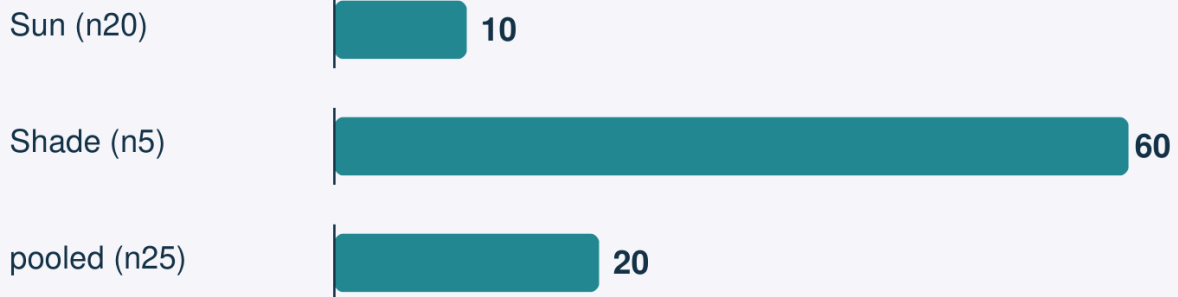
The facts we will use

Group Sun: 2 errors in 20 photos; Shade: 3 errors in 5. Same error definition; related plant views are documented.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Subgroup error rates

percent



Bar length shows the supplied value; the marked baseline is zero.

Subgroup Performance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Sun rate

Sun error= $2/20=10\%$.



2. Shade rate

Shade error= $3/5=60\%$.



3. Pooled rate

Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference.

CHANGE ONE THING

Shade gains 5 new cases with no errors.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Sun rate

What happens and why: Sun error= $2/20=10\%$.

Step: 2. Shade rate

What happens and why: Shade error= $3/5=60\%$.

Step: 3. Pooled rate

What happens and why: Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Small subgroup samples have high variability. An observed gap does not by itself establish discrimination or a causal explanation.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → **Diagnose** → **Repair**

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: A good overall result guarantees acceptable performance for every subgroup. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · **Expected** · **Actual** · **Likely cause** / **next test**

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE A good overall result guarantees acceptable performance for every subgroup.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Equal error rates alone do not establish fairness, and unequal rates alone do not identify discrimination or a cause. Groups may differ in base rates, label quality or conditions. Use results to ask better questions and plan review rather than label a group as inherently difficult.

Why does equal accuracy not settle whether use is fair?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Choose group definitions for a reason connected to use and possible harm. Retain error types, reference-label quality and sampling method. Avoid searching many groupings merely to find a dramatic or reassuring number. Report exploratory checks as exploratory.

Practise with less help: Check sensitivity

Use workbook W2 for the connected example “Subgroup evaluation”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Subgroup evaluation. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should the report retain subgroup rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Slice Audit: compute metrics and denominators for lighting, species, and camera; flag small samples and prioritize one gap.

1. Write the hypothesis and expected evidence before running.
2. Slice Audit: compute metrics and denominators for lighting, species, and camera; flag small samples and prioritize one gap.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: subgroup, performance gap, denominator, intersection.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A good overall result guarantees acceptable performance for every subgroup.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Add subgroup tables and a minimum acceptable threshold to the Plant Classifier release criteria.

THIS WEEK'S CHECKPOINT Add subgroup tables and a minimum acceptable threshold to the Plant Classifier release criteria.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF
Calculate subgroup accuracy and recall gaps with adequate denominators.

REQUIRED PROOF Calculate subgroup accuracy and recall gaps with adequate denominators.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Proxy Features and Fairness.

13. Study a complete case

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A has 18/20 correct; B has 6/10.

The question

Compute subgroup results and evidence limits.

Worked explanation

90% versus 60%, gap 30 percentage points. Report denominators and investigate collection/label differences without assuming an inherent group cause.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Report subgroup denominators with rates

Evidence or stage · Value or result

Evidence or stage: Group A

Value or result: 18/20

Evidence or stage: Group B

Value or result: 6/10

A has 18/20 correct; B has 6/10.

Compute subgroup results and evidence limits.

90% versus 60%, gap 30 percentage points. Report denominators and investigate collection/label differences without assuming an inherent group cause.

14. Try a new case independently

Independent case: Apply independently

Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 31 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Accuracy is 18/20 in bright light and 6/10 in dim light. Compute both rates and the percentage-point gap.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Add subgroup tables and a minimum acceptable threshold to the Plant Classifier release criteria.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the subgroup gap. What causal claim is not supported?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.